INCH-POUND

A-A-52417
January 7, 1993

COMMERCIAL ITEM DESCRIPTION

BATTERY, STORAGE: ALKALINE, NICKEL-CADMIUM

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-B-23272B and MIL-B-23272/1B, which are canceled.

1.0 Abstract. This CID covers the requirements for nickel-cadmium storage batteries primarily for service in military vehicles, starting of missile ground power equipment, radio operation, and as a source of electrical energy for operating vehicular accessories control mechanisms. The nickel-cadmium batteries have a wide tolerance to temperature extremes. They require less derating at low temperatures than lead acid units. Since the specific gravity of the battery does not change with the state of charge, they are not susceptible to freezing once a discharge is underway. Nickel-cadmium batteries can operate from plus 200 degrees Fahrenheit (°F) to minus 65°F without damage or without the threat of freezing.

1.1 Classification. Batteries shall be of the following types, as specified (see 5.2):

Type I - 2HNC 12-Volt, 35-ampere-hour capacity.
Type II - 6TNC 12-Volt, 70-ampere-hour capacity.

2.0 Salient characteristics.

2.1 Material. Material shall be as specified herein. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.7).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 6140

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2.1.1 Cell sealing compound. The cell sealing compound shall be alkali resistant, shall not shrink or separate from adjoining surfaces, and shall maintain an unbroken seal throughout an ambient temperature range from minus 65 degrees Fahrenheit (°F) to plus 200°F. The individual cells shall be sealed to maintain the seal when subjected to a pressure of two pounds per square inch (psi) throughout the ambient temperature range from minus 65°F to plus 200°F.

2.1.2 Electrolyte. The electrolyte used shall be a solution of potassium hydroxide and deionized or distilled water with a specific gravity of 1.300 ± 0.020 at plus 72°F. The electrolyte shall be furnished in the battery.

2.2 Design and construction. The battery shall be designed and constructed to the form and dimensions specified in (see figures 1 and 2).

2.2.1 Battery container. The battery container shall be of one piece, monobloc design. The container material shall be a high strength, alkali resistant, non-porous plastic which is resistant to temperatures ranging from minus 65°F to plus 200°F. The battery container shall conform to figures 1 and 2.

2.2.2 Cover. The cover shall be one piece high strength alkali resistant non-porous plastic. It shall insulate and protect all intercell connections enabling the battery to be submersible. There shall be no exposed intercell connections. The only exposed parts carrying current shall be the positive and negative terminal posts. The cover shall conform to figures 1, 2, and 3.

2.2.3 Terminal posts. Terminal posts shall conform to figure 3.

2.2.4 Corrugation. The sides of the external case shall be corrugated to conform to figure 4.

2.2.5 Handles. Handles shall be provided and shall be so designed to support the battery weight during lifting throughout a temperature range of minus 65°F to plus 200°F. Handles shall conform to figure 5.

2.2.5.1 Metal handles. Metal handles shall be of material that either by itself or when otherwise treated shall resist electrolyte (see 2.1.2). The handle shall show no corrosion or alkali absorption that interferes with performance of its functions. The handles shall operate freely in their retainers. Metal handles shall conform to figure 5.

2.2.5.2 Plastic handles. Plastic handles shall be rope type and if saturated with electrolyte conforming to 2.1.2, shall evidence no rupture, cracks, distortions, or other damage. Plastic handles shall conform to figure 5.

2.2.6 Vent plugs. Vent plugs shall be furnished for each cell and shall be located as shown in figures 1 and 2. The materials for vent plugs shall be non-absorbent and alkaline resistant and shall evidence no rupture, crack, distortions, or other damages. Plugs must be removable by hand without the use of any tool. Vent plugs shall have a gas or pressure release mechanism that is not affected or made inoperative by carbonate deposits. The vent plugs and their release or venting devices must be operative throughout the temperature range of minus 65°F to plus 250°F.

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2.3 Performance.

2.3.1 Assembled battery.

2.3.1.1 Ampere-hour capacity. The battery shall have an ampere-hour capacity not less than that specified in table I.

2.3.1.2 Cranking ability. The battery shall deliver the specified current for not less than the period of time specified in table I. During the first 5 seconds of cracking, the terminal voltage shall drop no lower than the value specified in table I.

TABLE I. Electrical performance characteristics.

Battery type	2HNC	6TNC
Number of cells	10	10
Weight filled	40 lbs	80 lbs
Nominal voltage	12 volts	12 volts
Rated capacity	35	70
5-hour rate	amp-hr	amp-hr
Cranking ability:		
Discharge rate	150 amps	300 amps
Minimum time at minus 40°F	5.5 minutes	5.5 minutes
Minimum time at plus 80°F	10.0 minutes	10.0 minutes
Minimum 5-second voltage at	8.0 volts	8.0 volts
minus 40°F	250	300
Life cycle minimum capacity		
Overcharge minimum capacity	9 cycles	14 cycles

2.3.1.3 Endurance. The battery shall complete not less than the number of cycles specified in table I. Complete discharging and shorting of the battery to balance the individual cell capabilities may be conducted when required by poor battery performance.

2.3.1.4 Overcharge cycle capacity. Overcharge cycle capacity shall be equal to 10 times the normal rated ampere-hour capacity of the battery as specified in table I. The open circuit phase shall be allowed for a maximum of 48 hours and the discharge phase rate shall be as specified in table I.

2.3.1.5 Retention of charge. Batteries shall show no more than 20 percent (%) decrease in rated five-hour capacity at the plus 80°F $\pm$ 5°F temperature capacity discharge and shall show no more than 80% decrease at the high rate as specified in table I at the minus 40°F temperature capacity discharge.

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2.3.1.6 Temperature rise. The battery while fully charged, shall be placed in an ambient temperature of 120° ± 5°F for a period of 24 hours. The battery shall then be discharged at the high rate specified in table I to a terminal voltage equivalent to an average of 0.6 volt per cell. Immediately following this discharge, the battery shall be placed across a potential of 15 volts and charged for a period of two hours and then discharged at the five-hour rate. The battery shall then satisfactorily meet 80 percent of the rated ampere-hour capacity as specified in table I.

2.3.2 Battery components.

2.3.2.1 Dielectric strength. The battery with all external surfaces and parts clean and dry shall be subjected to 2,500 volts root mean square (rms) at a frequency of 60 cycles per seconds (cps) between the current carrying parts and the container for 60 seconds. The container shall be free of leaks and other imperfections, and shall evidence no perforations or burning.

2.4 Physical.

2.4.1 Assembled battery.

2.4.1.1 Vibration resistance. The battery, while in a fully-charged state and at an ambient temperature of 80 ± 10°F, shall be subjected to a vibration consisting of a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion). The frequency shall be varied uniformly between the approximate limits of 10 and 55 cps and return to 10 cps and shall be traversed in approximately one minute. The battery shall maintain a steady voltage and current with no loosening of any component parts of the connectors, flooding of electrolyte at the top, broken connections, straps, plates or other damages which would result in a reduction of required performance.

2.4.1.2 Vent plug leakage. The vent plugs shall not leak and shall open to permit discharge of gas before the internal pressure reaches a maximum of 0.5 pounds per square inch cage (psig) in excess of the external pressure.

2.4.1.3 Terminal post torque resistance. The terminal posts shall withstand a torque of 250 pounds-inches applied in the direction perpendicular to the axis of the terminal posts without causing distortion, leakage, or other damage to the battery.

2.4.1.4 Dielectric strength. The battery container shall be free of leaks and other imperfections, and shall evidence no perforation, burning through, or breakdown of insulation (see 2.3.2.1).

2.5 Environmental.

2.5.1 Assembled battery.

2.5.1.1 Thermal shock resistance. The battery shall evidence no leakage or other damage which would result in a reduction of performance when subjected to the following:

a. Two psi internal pressure applied for two minutes with battery stabilized at an ambient temperature of plus 80°F ± 5°F.

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b. The pressure shall be removed. The battery shall be placed in an ambient temperature of minus 65°F ± 2°F for 24 hours or until the battery is stabilized then subjected to a two psi internal pressure for two minutes.

c. The pressure shall be removed and the battery shall immediately be placed in an ambient temperature of plus 190°F ± 5°F for 24 hours and then subjected to a two psi internal pressure for two minutes.

d. Cycle in accordance with subparagraph b & c shall then be repeated immediately.

2.5.2 Battery components.

2.5.2.1 Alkali absorption resistance. The cell container material shall not increase more than 1.5 percent in weight, a maximum of two percent in any dimension, and shall show no cracks or blisters. This shall be verified by immersing the battery in a covered vessel containing 150 cubic centimeter (cc) of alkaline solution of 1.300 specific gravity at 72°F and then placed in an oven for seven days at a temperature of 150° ± 4°F.

2.5.2.2 Corrosion resistance. All metal parts shall be subjected to a salt spray conforming to ASTM B117 for a continuous period of 200 hours. Thereafter they shall also be immersed in alkaline solution of 1.300 specific gravity at 72°F). The metal parts shall show no evidence of corrosion, and shall show no cracks or blisters.

2.6 Weight. The gross weight of the battery shall not exceed that specified in table I.

2.7 Markings. The battery markings shall be marked as specified on the applicable drawings. Marking shall include the following information in the order shown:

Battery, nickel cadmium
Type
Rated voltage
Rated capacity
National stock number
Manufacturer’s serial number
Manufacturer’s name
Date of manufacture
Contract or order number
U.S. Government property

2.7.1 Polarity markings. The polarity of the terminal posts shall be marked for identification. In lieu of the “POS” and “NEG”, the positive terminal may be identified by a “P” or a “+” sign, and the negative terminal post may be identified by an “N” or “-” sign. Marking shall be permanent and non-deteriorating.

2.7.2 Battery caution label. The battery shall be furnished with a caution label prominently located near the filler plugs. This label shall be in red printing and shall be resistant to electrolyte (see 2.1.2) and read as follows:

“CAUTION”
THIS BATTERY MUST BE SERVICED ONLY IN AN AUTHORIZED NICKEL-CADMIUM BATTERY SHOP.

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The word “CAUTION” shall be in .375 inch high minimum lettering. All other words shall be in 0.20 inch high lettering.

2.8 Instructions. Instructions for placing the battery into service shall be provided for each battery providing the following information: Care and maintenance; charge and discharge methods, rates and voltages; determining condition; electrolyte and safety precautions. The instructions shall be printed on a plate attached to each battery container on the side opposite the nameplate.

2.9 Material safety data sheet (MSDS). An MSDS shall be prepared in accordance with FED-STD-313 (see 5.2 and 5.8).

2.10 Identification and markings. Identification and markings for this alkaline nickel-cadmium battery shall be permanent and legible. The following information shall be included, as a minimum:

a. Manufacturer’s CAGE code.
b. Contract number.
c. Date of manufacture.
d. Part or identification number (PIN) (see 5.5).

2.11 Workmanship. Workmanship shall be such as to assure a product free of imperfect seals and bonds. Metallic parts shall be free of burrs, rust, scratches, chips, sharp edges, or other defects which will affect serviceability or appearance.

3.0 Quality Assurance Provisions.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer’s own drawing, specifications, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Government documents. FED-STD-313 “Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities”. Copies are available from the Navy Publications and Printing Service Office, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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5.1.2 Non-Government publications. ASTM B117 “Standard Test Method of Salt Spray (Fog) Testing; ASTM E380 “Standard Practice for Use of the International System of Units (SI) The Modernized Metric System” are available from the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Type of battery required and part number (see 1.1 and 5.5).
c. Selection of applicable level and packaging requirements (see 4.0).
d. Identify activities requiring copies of completed MSDS and specify when the MSDS will be inspected (see 2.9 and 5.6).

5.3 Metric product. Batteries that are manufactured to metric dimensions will be considered on the following basis:

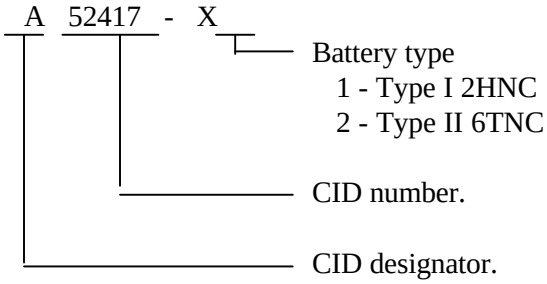
a. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of ASTM E380, and all other requirements of this CID are met.

b. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

c. The contracting officer has the option of accepting or rejecting the product.

5.4 International standardization agreements. Certain provisions (identified by *) of this standard are the subject of international standardization agreements ABC-ARMY-STD-140 and STANAG 4015. When revision or cancellation of this standard is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

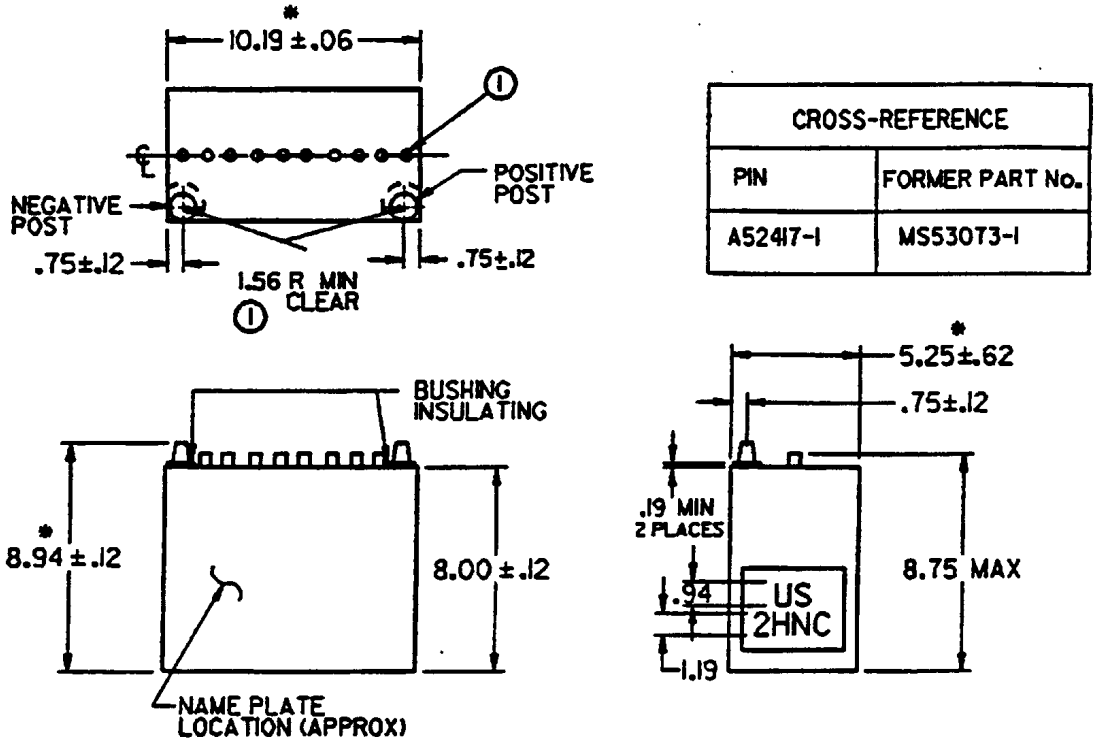
5.5 CID Based part identification numbers (PIN). The PINs to be used for batteries acquired to this CID are created as follows:



5.6 Cross-reference. Batteries conforming to this document are interchangeable/substitutable with batteries conforming to MIL-B-23272B.

5.7 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

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CROSS-REFERENCE	
PIN	FORMER PART No.
A52417-1	MS530T3-1

DIMENSIONS ARE IN INCHES

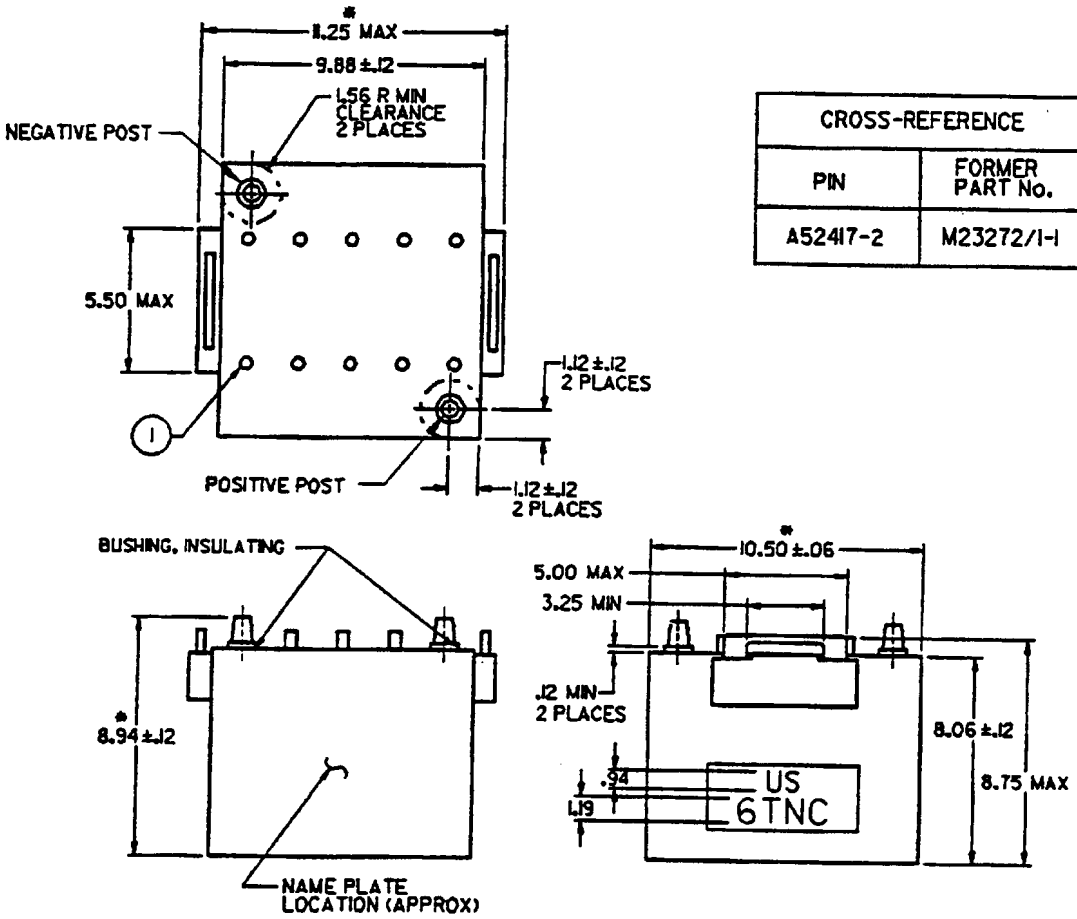
BATTERY TYPE	NUMBER OF CELLS	NOMINAL VOLTAGE	AMP HOUR CAPACITY 5 HR RATE	CRANKING ABILITY			LIFE CYCLE	OVERCHARGE
				MINIMUM TIME MINUTES		MINIMUM 5 SECOND VOLTS	MINIMUM CYCLE	MINIMUM CYCLE
				DISCHARGE RATE 150 AMPS				
				-40° F	+80° F	-40° F		
2HNC	10	12	35	5.5	10.0	8.0	250	9

NOTES:

1. Vent plugs must be removable without use of any tool or device.
2. Plug locations shall be fixed only in respect to required minimum distance from terminal posts and sides of battery.
3. Items flagged with an asterisk (*) are subject to International Standardization Agreements (see 5.4).

FIGURE 1. Battery, storage, nickel-cadmium (type I).

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CROSS-REFERENCE	
PIN	FORMER PART No.
A52417-2	M23272/1-1

DIMENSIONS ARE IN INCHES

BATTERY TYPE	NUMBER OF CELLS	NOMINAL VOLTAGE	AMP HOUR CAPACITY 5 HR RATE	CRANKING ABILITY			LIFE CYCLE	OVERCHARGE
				MINIMUM TIME MINUTES		MINIMUM 5 SECOND VOLTS	MINIMUM CYCLE	MINIMUM CYCLE
				DISCHARGE RATE 300 AMPS				
				-40° F	+80° F	-40° F		
6TNC	10	12	70	5.5	10.0	8.0	300	14

NOTES:

1. Vent plugs must be removable without use of any tool or device.
2. Plug locations shall be fixed only in respect to required minimum distance from terminal posts and sides of battery.
3. Items flagged with an asterisk (*) are subjected to International Standardization Agreements (see 5.4).

FIGURE 2. Battery, storage, nickel-cadmium (type II).

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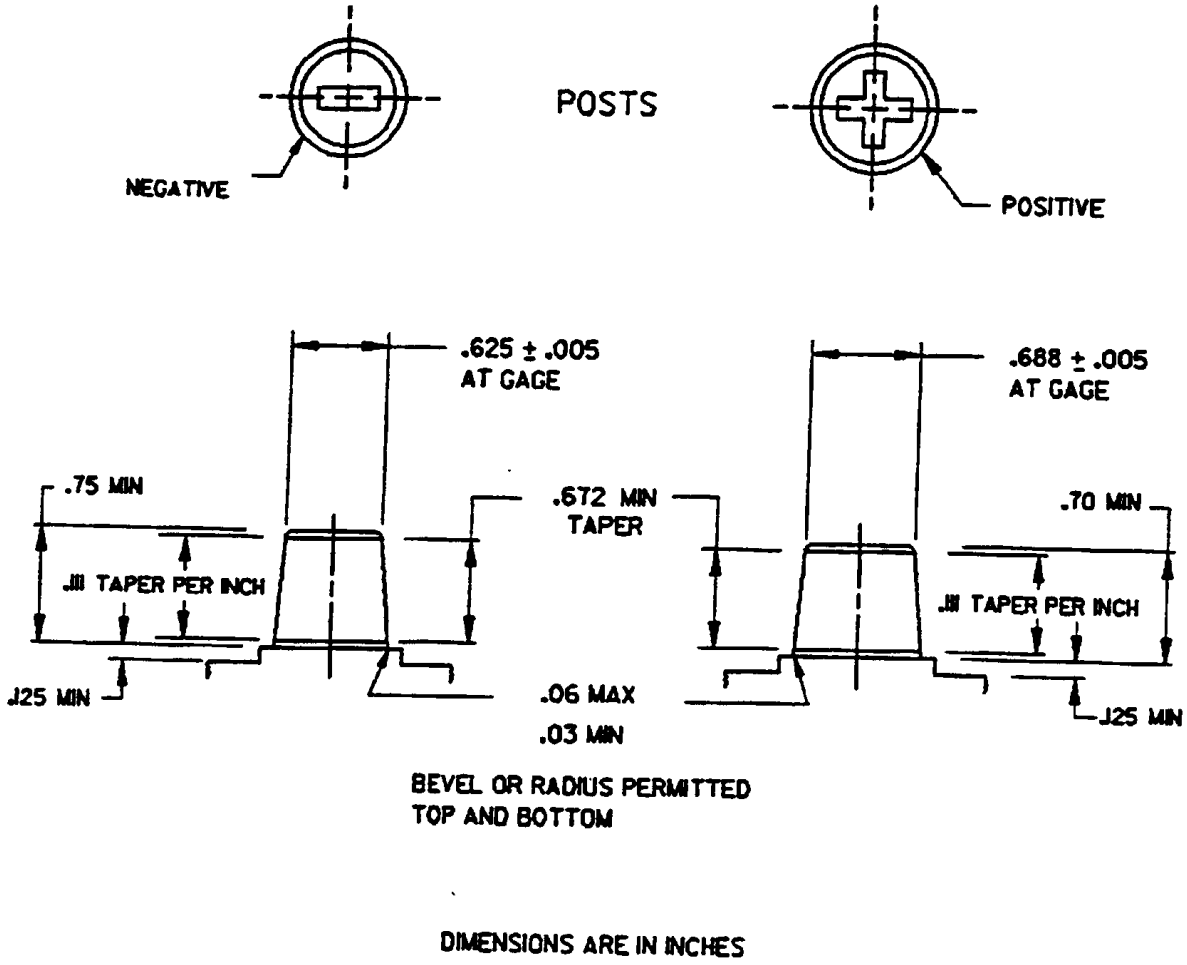


FIGURE 3. Terminal posts.

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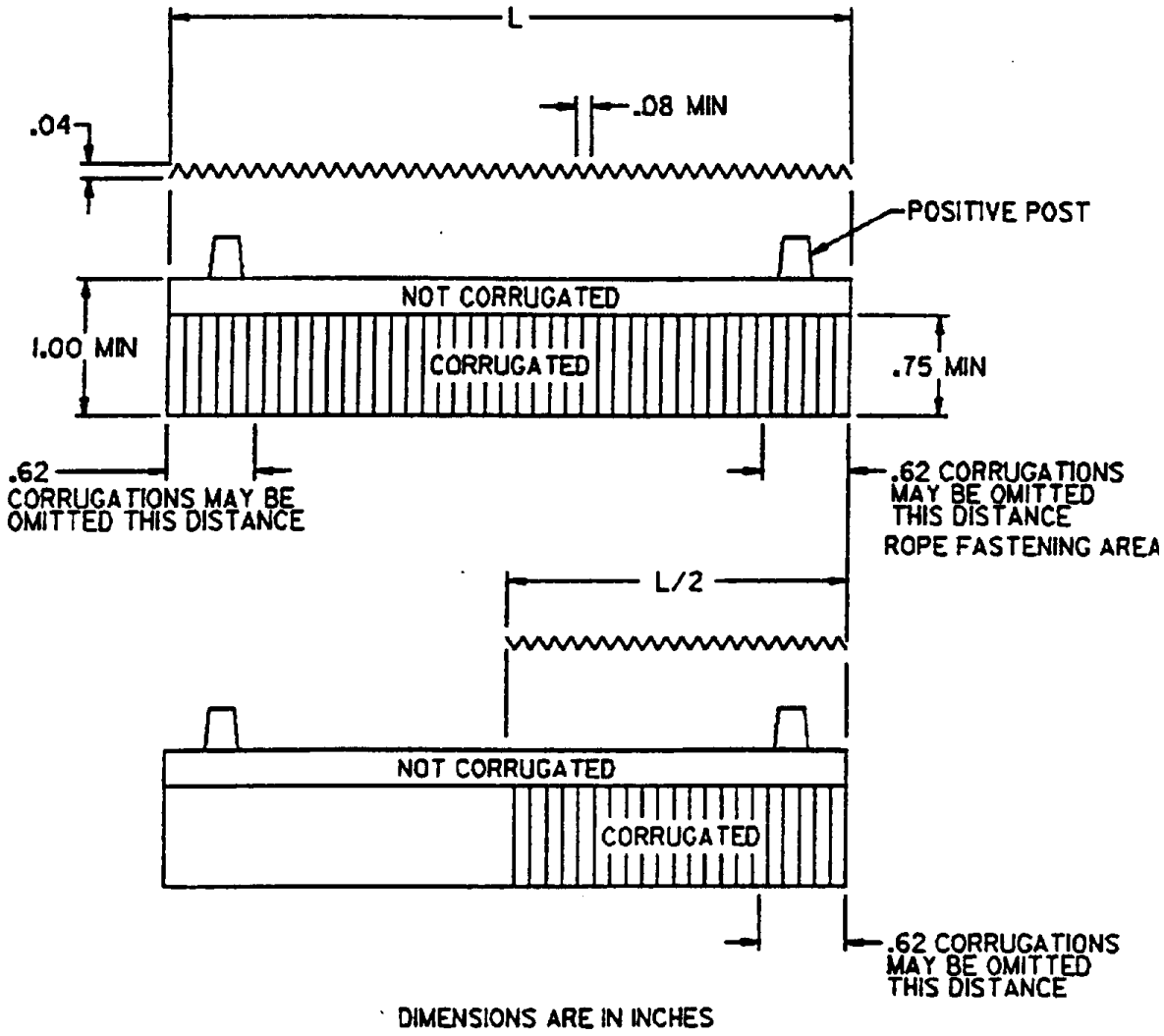
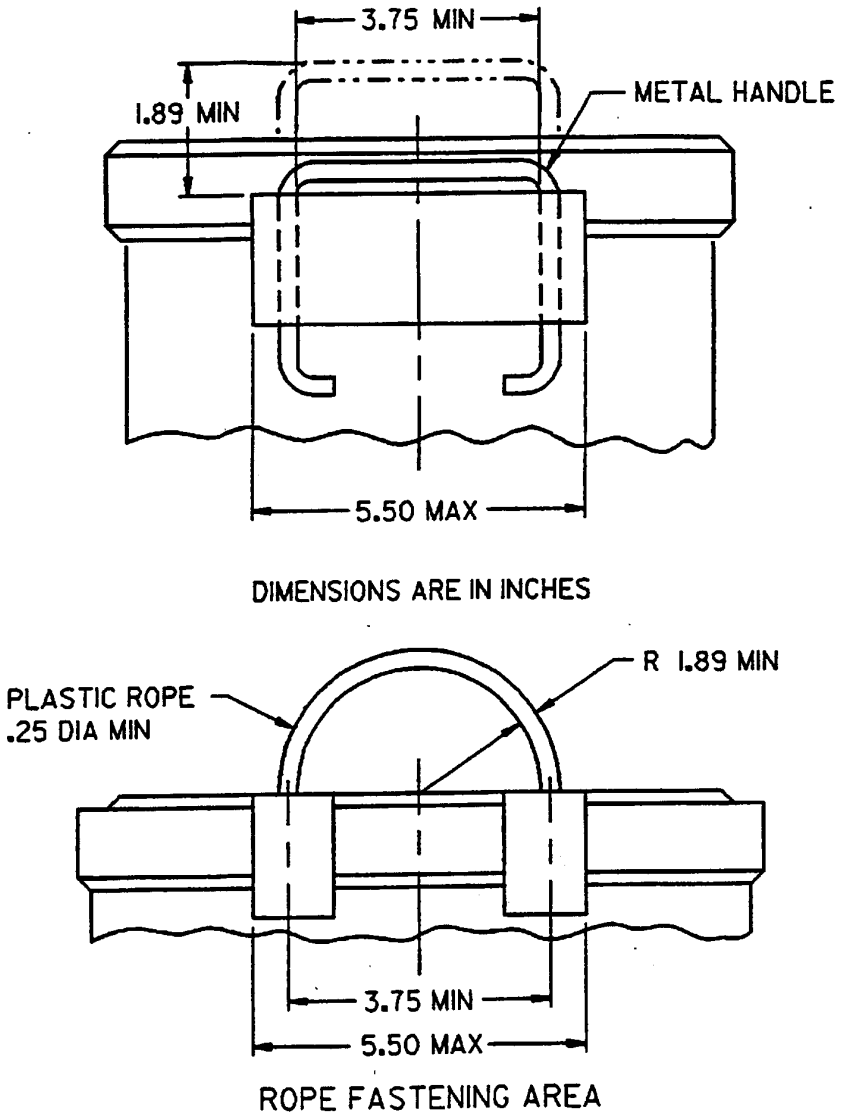


FIGURE 4. Cover corrugation.

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NOTE: THE SPECIFIC DESIGN, CONSTRUCTION, AND CONFIGURATION OF THE HANDLE IS OPTIONAL PENDING THE REQUIREMENTS SPECIFIED HERIN ARE MET (SEE 2.2.5).

(Plastic rope construction optional)
FIGURE 5. Handles.

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5.8 MSDS. The contracting officer should identify those activities requiring copies of the completed MSDS prepared in accordance with FED-STD-313. Additional pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313 (see 2.9).

Custodians:
Army - AT
Air Force - 99

User Activities:
Army - EL, ME

Review Activities:
Air Force - 82
DLA - GS

Preparing Activity:
Army - AT

(Project 6140-0761)

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